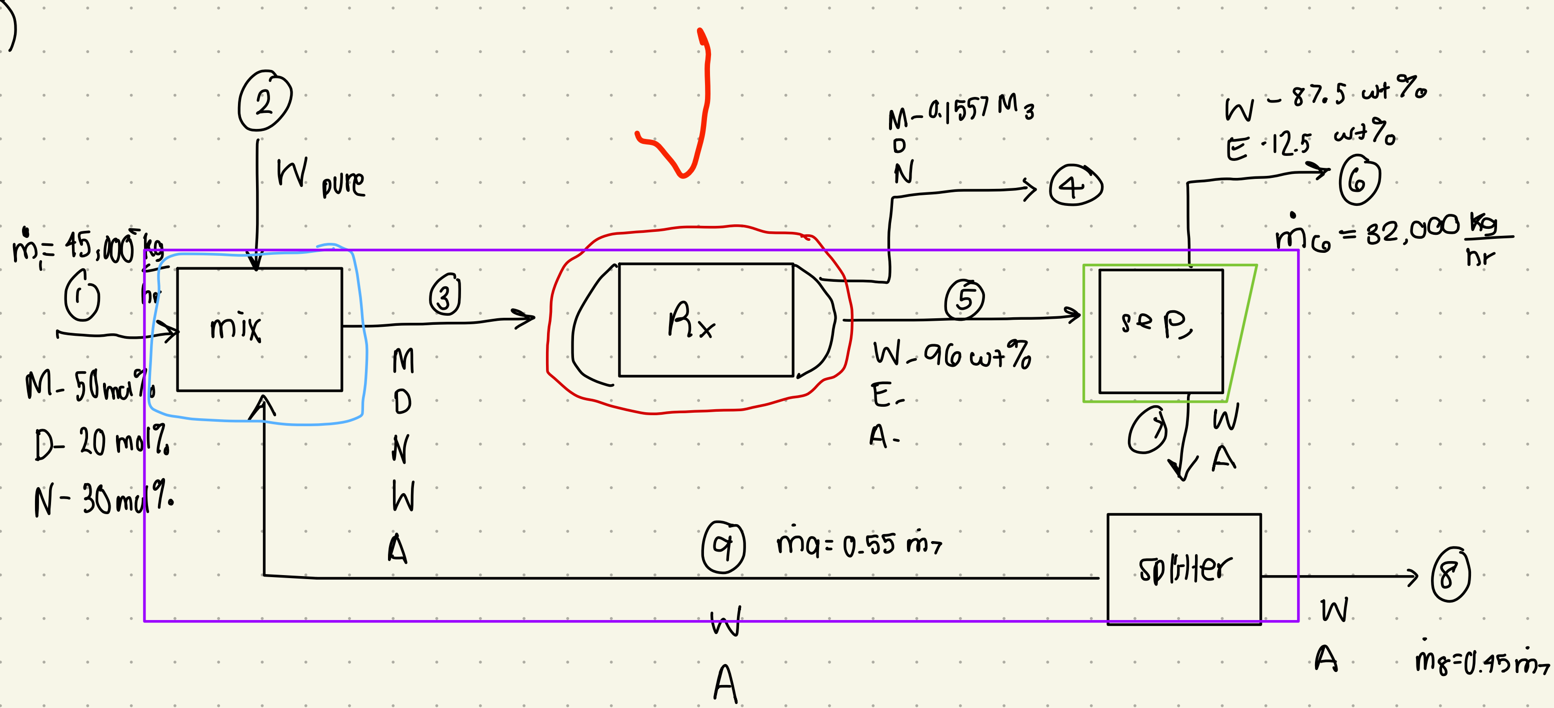
HW 8

(1)



Total DOF: 4a) 4b) 5a) 5b) 5c) 5d)

$23 + 2 - 2 - (4 + ((2-1)(2-1)) - 2 - 16 = 0$ well posed

40/40

mixer balance:

$MW_{CO} = 28\ g/mol$

$MW_{CO_2} = 44\ g/mol$

$MW_{N_2} = 28\ g/mol$

$MW_W = 18\ g/mol$

$MW_{CH_3COOH} = 60\ g/mol$

1 mole tot: $M = 0.5\ \text{mole} \cdot \frac{28g}{mol} = 14\ g$

$D = 0.2\ mol \cdot \frac{44g}{mol} = 8.8\ g$

$N = 0.3\ mol \cdot \frac{28g}{mol} = 8.4\ g$

tot $\frac{g}{mole} = 31.2 \frac{g}{mol}$

$\dot{n}_1 = \frac{45000\ kg/hr}{31.2\ g/mol}$

$\dot{n}_1 = 1442.3 \frac{kmol}{hr}$

if $\dot{n}_1 = 1442.3\ kmol/hr$

$M_1 = 721.15\ kmol\ M/hr$

$D_1 = 288.46\ kmol\ D/hr$

$N_1 = 432.69\ kmol/hr$

Species balance

$M_3 = M_1 = 721.15\ kmol\ M/hr$

$D_3 = D_1 = 288.46 \frac{kmol}{hr} D$

$N_3 = N_1 = 432.69 \frac{kmol}{hr} N$

$W_3 = W_2 + W_9$

$A_3 = A_9 - 26.62 \frac{kmol}{hr}$

Reactor Balance:

$M_4 = M_3 - 6\xi_1 - 4\xi_2 = 0.1557 M_3 \Rightarrow M_4 = 721.15 - 6\xi_1 - 4\xi_2 = 0.1557(721.15)$ kmol/hr

$D_4 = D_3 + 4\xi_1 + 2\xi_2 \Rightarrow D_4 = 288.46 + 4\xi_1 + 2\xi_2$

$N_4 = N_3 \Rightarrow N_4 = 432.69$ kmol N/hr

$A_5 = A_3 + \xi_2 \Rightarrow A_5 = A_9 + \xi_2 \Rightarrow A_9 = 48.4 - 21.78 = 26.62$ kmol/hr

$E_5 = \xi_1 \Rightarrow E_5 = \xi_1 = 86.96$ kmol/hr

$W_5 = W_3 - 3\xi_1 - 2\xi_2 \Rightarrow W_5 = W_2 + W_9 - 3\xi_1 - 2\xi_2$

$M_4 = 112.28 \frac{kmol}{hr} = 721.15 \frac{kmol}{hr} - 6(86.96)\frac{kmol}{hr} - 4\xi_2 \Rightarrow \xi_2 = 21.78 \frac{kmol}{hr}$

$D_4 = 288.46 + 4(86.96) + 2(21.78) = 679.86$ kmol/hr

Seperator Balance:

$W_5 = W_6 + W_7$

$E_5 = E_6 = 86.96$ kmol E/hr

$A_5 = A_7 = 48.4$ kmol A/hr

$E_6 = 0.125 \cdot 32{,}000 \frac{kg}{h} = 4000$ kg/h

$MW_E = 46$ g/mol

$E_6 = 4000 \frac{kg}{hr} \cdot \frac{mol}{46g} = 86.96 \frac{kmol\ E}{hr}$

$W_6 = 0.875\left(32000 \frac{kg}{h}\right)\cdot\left(\frac{mol}{18g}\right) = 1555.56 \frac{kmol\ W}{hr}$

$\dot{n}_6 = 1555.56 \frac{kmol\ W}{hr} + 86.96 \frac{kmol\ E}{hr} = 1642.51 \frac{kmol}{hr}$

overall balance:

$$\left.\begin{aligned} M_4 &= M_1 - 6\xi_1 - 4\xi_2 \\ D_4 &= D_1 + 4\xi_1 + 2\xi_2 \\ N_1 &= N_4 \end{aligned}\right\} \text{redundant}$$

$3\xi_1 + 2\xi_2 = W_2 - W_6 - W_8 \Rightarrow W_2 = 3\xi_1 + 2\xi_2 + W_6 + W_8$

$A_8 = \xi_2 = 21.78 \frac{kmol}{hr}$ $\quad 0.45 A_7 = A_8 \rightarrow A_7 = 21.78/0.45 = 48.4$ kmol A/hr

Calculations

(1) $\dot{n}_1 = 1442.3$ Kmol/hr

$M_1 = 721.15$ Kmol M/hr

$D_1 = 288.46$ Kmol D/hr

$N_1 = 432.69$ Kmol/hr

(2) $\dot{n}_2 = W_2 = W_6 + W_8 + 304.44$

$W_2 = 1555.56 + 3448.43 + 304.44$

$W_2 = 5308.43$ Kmol $\frac{W}{hr}$

(3) $\dot{n}_3 = 10992.1$ Kmol/hr

$M_3 = 721.15$ Kmol/hr

$D_3 = 288.46$ Kmol D/hr

$N_3 = 432.69$ Kmol N/hr

$A_3 = 26.62$ Kmol A/hr

$W_3 = W_2 + W_9 = 5308.43 + 4214.75 = 9523.18$ $\frac{Kmol}{hr}$

(4) $\dot{n}_4 = 1224.83$ kmole/hr

$M_4 = 112.28$ Kmol $\frac{M}{hr}$

$D_4 = 679.86$ Kmol $\frac{D}{hr}$

$N_4 = 432.69$ Kmol $\frac{N}{hr}$

(5) $\dot{n}_5 = 9354.11$ Kmol/hr

$W_5 = 0.96\,\dot{m}_5 \cdot \frac{mol}{18g} = 9218.75$ Kmol $\frac{W}{hr}$

$E_5 - 86.96$ Kmol E/hr

$A_5 - 48.4$ Kmol A/hr

$\dot{m}_5 = \dot{n}_5 \cdot MW_{avg} = \frac{6904.16\,kg}{hr} + W5\left(\frac{18g}{mol}\right)$

$MW_{avg} = \frac{48.4\,\frac{kmol}{hr}}{\dot{n}_5} \cdot \frac{60g}{mol} + \frac{86.96\,\frac{kmol}{hr}}{\dot{n}_5} \cdot \frac{46g}{mol} + \frac{W_5}{\dot{n}_5}\left(\frac{18g}{mol}\right)$

$\dot{m}_5 = \left(6904.16\,\frac{kg}{hr} + W_5\left(\frac{18g}{mol}\right)\right)$

$W_5 = \left(6904.16\,\frac{kg}{hr} + W_5\left(\frac{18g}{mol}\right)\right) \cdot \frac{mol}{18g} \cdot 0.96$

$0.04\,W_5 = 368.75$ $\frac{Kmol}{hr}$

$W_5 = 9218.75$ Kmol $\frac{W}{hr}$

(6) $\dot{n}_6 = 1642.52$ Kmol/hr

$W_6 = 1555.56$ Kmol W/hr

$E_6 = 86.96$ Kmol E/hr

(7) $\dot{n}_7 =$

$W_7 = W_5 - 1555.56$ Kmol W/hr

$= 9218.75 - 1555.56 = 7663.19$ $\frac{Kmol}{hr}$

$A_7 = 48.4$ Kmol $\frac{A}{hr}$

(8) $\dot{n}_8 = 3470.21$ Kmol/hr

$A_8 - 21.78$ Kmol/hr

$W_8 - 0.45\,W_7 = 3448.43$ Kmol $\frac{W}{hr}$

(9) $\dot{n}_9 = 4241.37$

$A_9 - 26.62$ Kmol $\frac{A}{hr}$

$W_9 = 0.55\,W_7$

$= 4214.75$ Kmol $\frac{W}{hr}$

Stream 1 $\dot{n}_1 = 1442.3$ Kmol/hr

$M_1 = 721.15$ Kmol M/hr

$D_1 = 288.46$ Kmol D/hr

$N_1 = 432.69$ Kmol/hr

Stream 5 $\dot{n}_5 = 9354.11$ Kmol/hr

$W_5 = 9218.75 \frac{\text{Kmol W}}{\text{hr}}$

$E_5 = 86.96$ Kmol E/hr

$A_5 = 48.4$ Kmol A/hr

Stream 2 $\dot{n}_2 = W_2 = 5308.43 \frac{\text{Kmol W}}{\text{hr}}$

Stream 6 $\dot{n}_6 = 1642.52$ Kmol/hr

$W_6 = 1555.56$ Kmol W/hr

$E_6 = 86.96$ Kmol E/hr

Stream 3 $\dot{n}_3 = 10992.1$ Kmol/hr

$M_3 = 721.15$ Kmol/hr

$D_3 = 288.46$ Kmol D/hr

$N_3 = 432.69$ Kmol N/hr

$A_3 = 26.62$ Kmol A/hr

$W_3 = 9523.18 \frac{\text{Kmol W}}{\text{hr}}$

Stream 7 $\dot{n}_7 = 7711.59$ Kmol/hr

$W_7 = 7663.19 \frac{\text{Kmol}}{\text{hr}}$

$A_7 = 48.4 \frac{\text{Kmol A}}{\text{hr}}$

Stream 4 $\dot{n}_4 = 1224.83$ kmole/hr

$M_4 = 112.28 \frac{\text{Kmol M}}{\text{hr}}$

$D_4 = 679.86 \frac{\text{Kmol D}}{\text{hr}}$

$N_4 = 432.69 \frac{\text{Kmol N}}{\text{hr}}$

Stream 8 $\dot{n}_8 = 3470.21$ Kmol/hr

$A_8 - 21.78$ Kmol/hr

$W_8 = 3448.43 \frac{\text{Kmol W}}{\text{hr}}$

Stream 9 $\dot{n}_9 = 4241.37$ Kmol/hr

$A_9 - 26.62 \frac{\text{Kmol A}}{\text{hr}}$

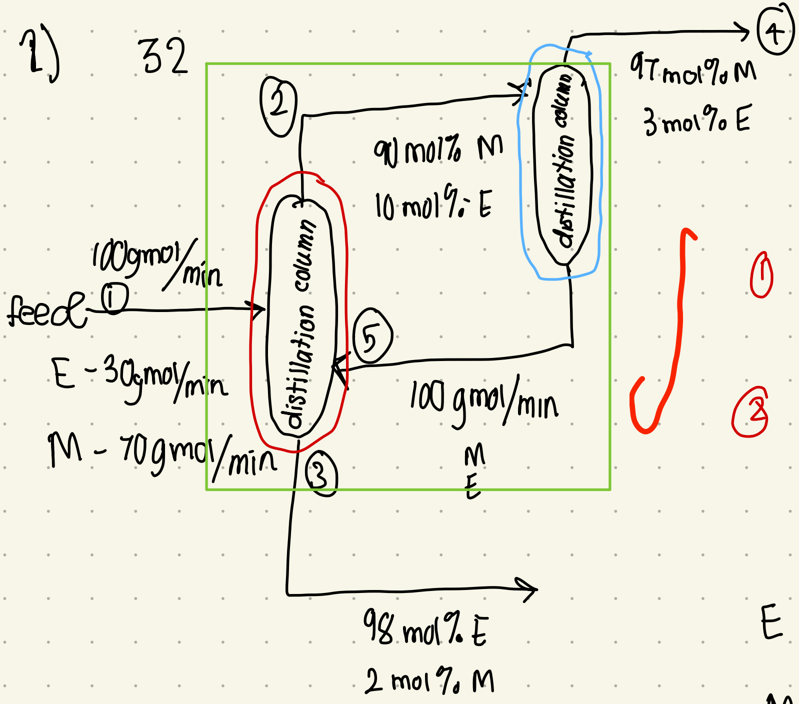
$W_9 = 4214.75 \frac{\text{Kmol W}}{\text{hr}}$

c) single pass conversion $= \dfrac{\left(9{,}523.18 \frac{\text{Kmol}}{\text{hr}} - 9218.75 \frac{\text{Kmol}}{\text{hr}}\right)}{9523.18 \frac{\text{Kmol}}{\text{hr}}} \times 100\% = \boxed{3.196\%}$

overall conversion $= \dfrac{\left(5308.43 \frac{\text{Kmol}}{\text{hr}} - (1555.56 + 3448.48) \frac{\text{Kmol}}{\text{hr}}\right)}{5308.43 \frac{\text{Kmol}}{\text{hr}}} = \boxed{5.73\%}$

Adding more co would allow for more efficient use of water in the system.

2) 32



Distillation Column 1

① E: $30\frac{gmol}{min} + x_{E5}(100)\frac{gmol}{min} = 0.1\dot{n}_2 + 0.98\dot{n}_3$

② M: $70\frac{gmol}{min} + x_{M5}\left(100\frac{gmol}{min}\right) = 0.9\dot{n}_2 + 0.02\dot{n}_4$

③ $\dot{n}_1 + \dot{n}_5 = \dot{n}_2 + \dot{n}_3$

Column 2

E: $0.1\dot{n}_2 = 0.03\dot{n}_4 + 100x_{E5}$

M: $0.9\dot{n}_2 = 0.97\dot{n}_4 + 100x_{M5}$

total

④ E: $30\frac{gmol}{min} = 0.98\dot{n}_3 + 0.03\dot{n}_4$

5 eqn
unk

30/35

⑤ M: $70\frac{gmol}{min} = 0.97\dot{n}_4 + 0.02\dot{n}_3$

⑤ $\dot{n}_3 = 3500 - 48.5\dot{n}_4 = 28.85\frac{gmol}{min}$

④ $30 = 3430 - 47.53\dot{n}_4 + 0.03\dot{n}_4$

$\dot{n}_4 = 71.57\frac{gmol}{min}$

③ $\dot{n}_2 = 100\frac{gmol}{min} + 100\frac{gmol}{min} - 28.85 = 171.15\frac{gmol}{min}$

① $x_{E5} = \left(\frac{0.1(171.15) + 0.98(28.85) - 30}{100}\right)\frac{gmol}{min} = 0.15388 \approx 0.15$

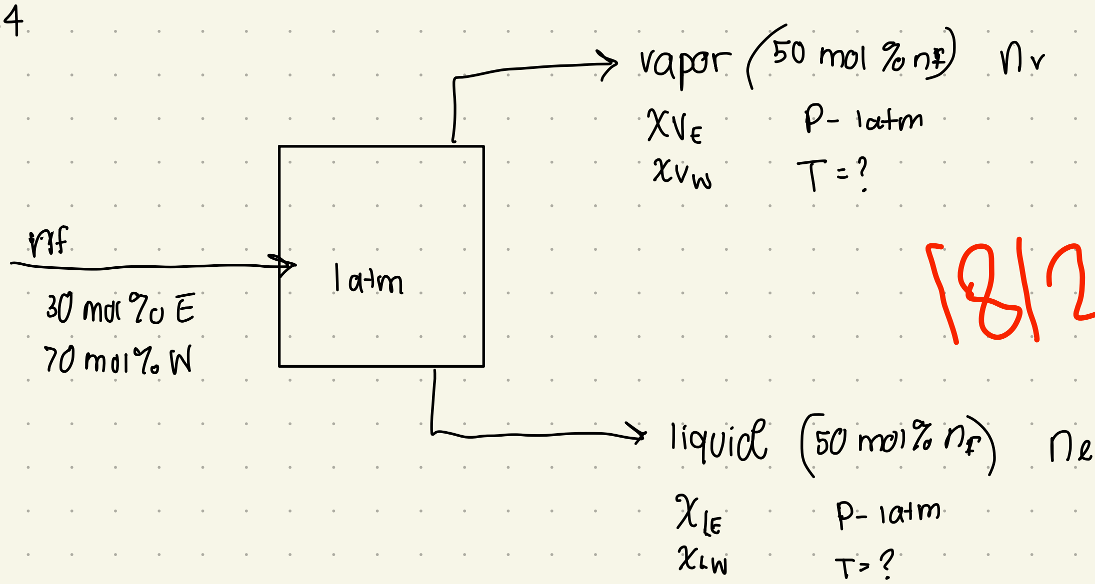
$x_{M5} = 0.84612 \approx 0.85$

-5 pts

$$\boxed{\begin{aligned}\dot{n}_4 &= 71.57\frac{gmol}{min}\\ x_{E5} &= 0.15\\ x_{M5} &= 0.86\end{aligned}}$$

Stream 4 fractional recovery	Stream 3 recovery
$f_E = 0.03(71.57)/0.1(171.15) \cdot 100\% = \boxed{1.25\%}$	$f_E = 0.98(28.85)/30 + 100(0.15) \cdot 100 = \boxed{62.8\%}$
$f_M = 0.97(71.57)/0.9(171.15) \cdot 100\% = \boxed{45\%}$	$f_M = \frac{0.02(28.85)}{70 + 100(0.85)} = \boxed{0.372\%}$

(3) P5.34



18/25

P = 760 mmHg

$$\log_{10} P_E^{sat} = 8.04494 - \frac{1554.3}{T + 222.65} \Rightarrow P_E^{sat} = 10^{8.04494} - 10^{\frac{1554.3}{T+222.65}}$$

$\frac{V}{F} = \frac{50}{100} = 0.5$

$$\log_{10} P_W^{sat} = 8.10765 - \frac{1750.286}{T + 235} \quad (0-60\,°C)$$

$$P_W^{sat} = 10^{8.10765} - 10^{\frac{1750.286}{T+235}}$$

mol balance

basis $n_f = 100$ mol

$n_f = 100$ mol
$n_v = 50$ mol
$n_\ell = 50$ mol

$n_f = n_v + n_\ell$

$30\ \cancel{mol}\ E = 50 x_{VE} + 50 x_{LE}$

70 mol $W = 50 x_{VW} + 50 x_{LW}$

$K_i = \frac{y_i}{x_i}$ $\quad K_i = \frac{P_{sat}}{P}$

$y_i = \frac{P_{sat}}{P} x_i$

$$0 = \sum \frac{z_i(K_i - 1)}{1 + \frac{V}{F}(K_i - 1)}$$

T: 0-60 :

$$0 = \frac{0.3\left(\frac{10^{8.04494} - 10^{\frac{1554.3}{T+222.65}}}{760} - 1\right)}{1 + 0.5\left(\frac{10^{8.04494} - 10^{\frac{1554.3}{T+222.65}}}{760} - 1\right)} + \frac{0.7\left(\frac{10^{8.10765} - 10^{\frac{1750.286}{T+235}}}{760} - 1\right)}{1 + 0.5\left(\frac{10^{8.10765} - 10^{\frac{1750.286}{T+235}}}{760} - 1\right)}$$

using goal seek T = 93.39 °C

X -1 pts

at $T = 93.39$

liquid

$$m = \left(\frac{0.019 - 0.0721}{95.5 - 89}\right) = -0.008169$$

$$0.019 = -0.008169(95.5) + b \Rightarrow b = 0.97016$$

$$x_{LE} = -0.008169(93.39) + 0.97016 = 0.2072$$

so $x_{LW} = 1 - 0.2072 = 0.7928$ -2 pt

vapor

$$m = \left(\frac{0.17 - 0.3891}{95.5 - 89}\right) = -0.0337$$

Product compositions

$$0.17 = -0.0337(95.5) + b \rightarrow b = 3.389$$

$$x_{VE} = -0.0337(93.39) + 3.389 = 0.241$$

$$x_{VW} = 0.759$$

$x_{LE} - 0.2072$
$x_{LW} - 0.7928$
$x_{VE} - 0.241$
$x_{VW} - 0.759$

fractional recovery

liquid

$$f_E = \frac{0.2072(50)}{30} \cdot 100\% = \boxed{34.53\% = f_{LE}}$$

$$f_W = \frac{0.7928(50)}{70} \cdot 100\% = \boxed{56.62\% = f_{LW}}$$

vapor

$$f_E = \frac{0.241(50)}{30} \cdot 100\% = \boxed{40.16\% = f_{VE}}$$ -2 pts

$$f_W = \frac{0.759(50)}{70} \cdot 100\% = \boxed{54.2\% = f_{VW}}$$

separation factor : $\alpha = \left(0.241/0.759\right) \Big/ \left(\frac{0.2072}{0.7928}\right) = \boxed{1.2149}$

where A = ethanol
B = water

-2 pts